

PROJECT REVIEW-3

Event and Room Occupancy Management System

Team 1 | Section 10

Course: Database Systems Engineering and Distributed Backend Development (25CS1302E)

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Project Overview

What it does

A web system to create, schedule and track events and the rooms they occupy.

Why it is needed

Manual scheduling causes room clashes, poor utilisation and scattered records.

Main functionality

Event and room management, event–room assignment, occupancy view.

Technologies

React.js • Node.js + Express.js • MySQL • REST APIs

Front-End Development

Built with React.js

Component-based UI

Reusable components

Navigation

Page-to-page routing

Forms

Add / edit inputs

Data display

Tables and lists

User interaction

Clicks, filters, actions

Dynamic rendering

UI updates from state

API integration

REST calls to backend

Frontend Modules / Features

Dashboard

Event Management

Room Management

**Event-Room
Assignment**

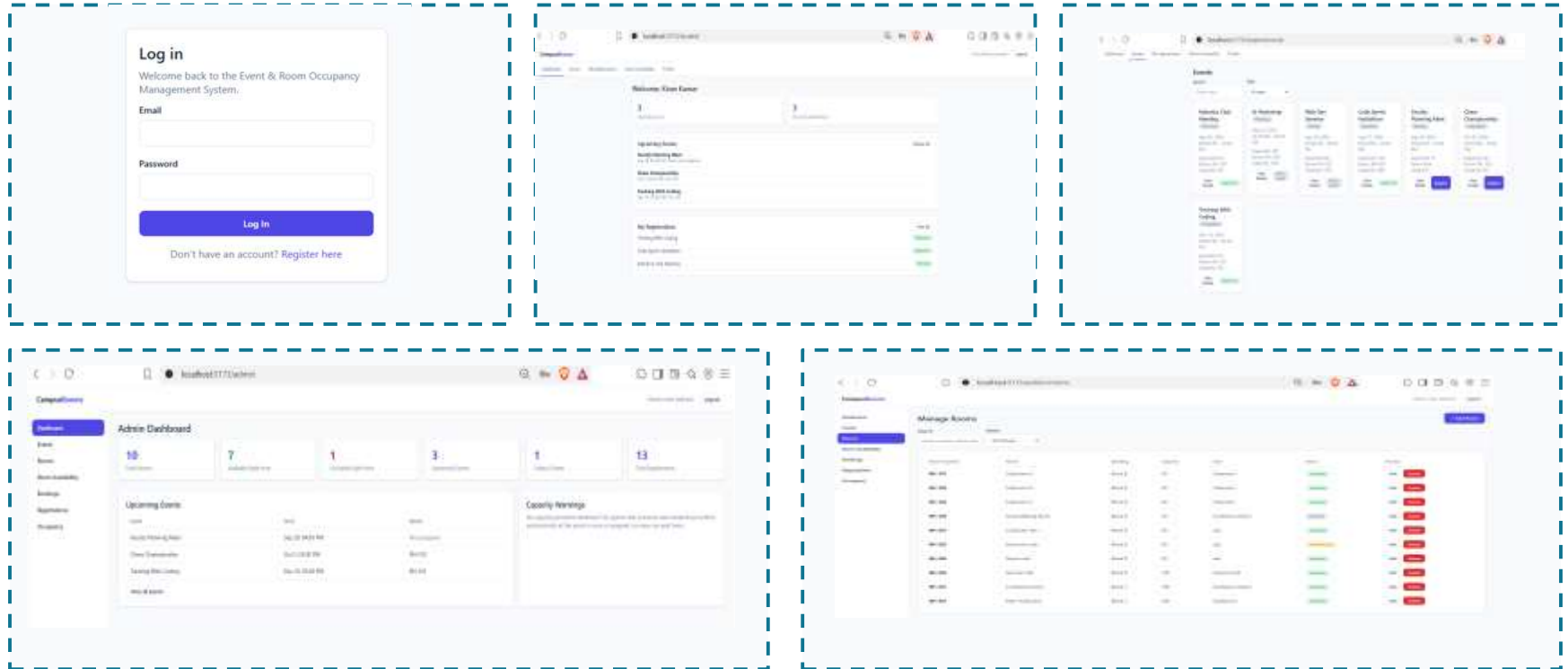
Occupancy View

Add/Edit Forms

Data Tables

**Validation &
Feedback**

Frontend Screenshots



Frontend–Backend–Database Flow



- User actions in React trigger REST API requests (GET, POST, PUT, DELETE).
- Express routes handle the request and run SQL queries on MySQL.
- The database result returns as a JSON response.
- React updates its state and re-renders the data on screen.

Database Design

event_room_management

Relational approach

Data stored in linked tables

Purpose

Store events, rooms and occupancy data

Data consistency

Keys and constraints enforce valid data

Reduced redundancy

Each fact stored once, then referenced

Relationships

Entities connected through foreign keys

ER Diagram / Database Schema

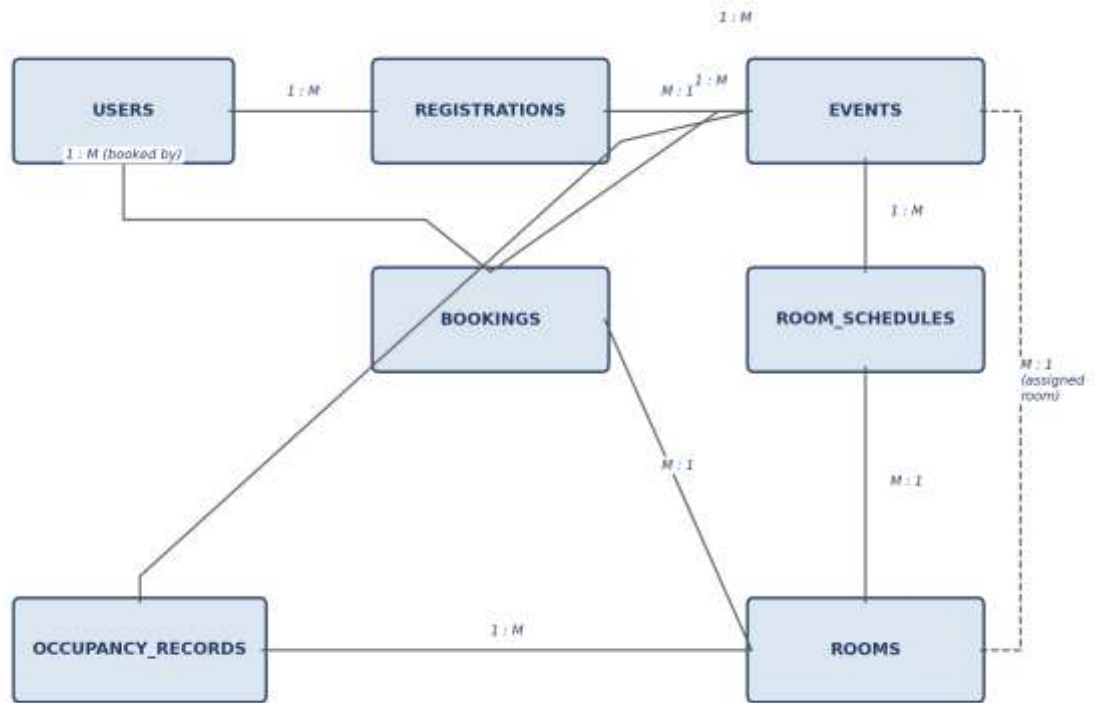


Figure 5.1: Entity-Relationship diagram of the 7-table database schema

Tables and Their Purpose

Table	Purpose
users	Stores student and admin account information
rooms	Stores room details, status, and capacity
events	Stores event details, scheduling, and room assignment
registrations	Links users/students with events
room_schedules	Stores room schedules and supports room conflict checking
bookings	Records room/event reservations and booking details
occupancy_records	Stores room occupancy measurements for events

Keys and Relationships

Primary Key

Uniquely identifies each row in a table.

Foreign Key

Column that references the primary key of another table.

Referential Integrity

Prevents records pointing to rows that do not exist.

Why it matters

Relationships avoid duplicate data and keep tables consistent.

Database Operations (CRUD)

CREATE

Add events, rooms
and related records

READ

View event, room and
occupancy
information

UPDATE

Edit event, room and
assignment details

DELETE

Remove records when
required

Sample SQL Queries

Replace placeholders with actual table and column names from the schema

1. View all events

```
SELECT * FROM <events_table>;
```

2. View all rooms

```
SELECT * FROM <rooms_table>;
```

3. Events with their rooms (JOIN)

```
SELECT e.<event_name>, r.<room_name>  
FROM <events_table> e  
JOIN <assignment_table> a ON a.<event_id> = e.<event_id>  
JOIN <rooms_table> r ON r.<room_id> = a.<room_id>;
```

Current Implementation / Progress

Front-End Development

- React frontend implemented
- Major UI modules developed
- Forms and data views implemented
- API integration completed/tested
- Dynamic data displayed

Database Design

- MySQL database created
- 7 relational tables implemented
- Primary/foreign key relationships established
- Sample data populated
- CRUD operations tested

Conclusion and Next Steps

The system combines a React frontend with a relational MySQL database to manage events, rooms and occupancy information.

Next Steps

- UI/UX improvements
- Additional validation
- Better error handling
- More testing
- Occupancy management improvements

THANK YOU